

Decode Labs Project 2

Chain-of-Thought Prompt Engineering

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1: Objective

To design a Chain-of-Thought (CoT) prompt that forces an AI model to solve multi-step reasoning problems systematically while reducing hallucinations and improving accuracy through self-correction.

2: Problem Statement

Large Language Models often provide answers without showing their reasoning process, leading to mistakes in logic and calculations. This project aims to create a prompt that guides the model through structured reasoning, verification, and final answer generation.

3: Build Main Prompt

You are a logical reasoning engine.

Instructions:

- 1. Read the problem carefully.*
- 2. Break the problem into smaller steps.*
- 3. Solve each step individually.*
- 4. Verify each calculation.*
- 5. Perform a self-correction review.*
- 6. If an error is found, correct it.*
- 7. Provide the final answer only after verification.*

Output Format:

Step 1: Understanding

Step 2: Reasoning

Step 3: Calculation

Step 4: Self-Check

Final Answer:

4: Test Cases

TEST CASE 1:

INPUT:

*If it takes 5 machines 5 minutes to make 5 widgets,
How long will it take 100 machines to make 100 widgets?*

EXPECTED OUTPUT:

"5 minutes"

TEST CASE 2:

INPUT:

*A bat and ball cost \$1.10.
The bat costs \$1 more than the ball.
How much does the ball cost?*

EXPECTED OUTPUT:

"\$0.05"

TEST CASE 3:

INPUT:

*Lily pads double every day.
A lake becomes fully covered in 48 days.
When was it half covered?*

OUTPUT:

"47 days"

5: PROMPT TESTING SCREENSHOTS

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Sure. Please provide the problem you want me to solve, and I'll use this format:

Step 1: Understanding

Step 2: Reasoning

Step 3: Calculation

Step 4: Self-Check

Final Answer:



Screenshot 1: *Machines Problem*

You are a logical reasoning engine.

[Prompt]

Question:

If it takes 5 machines 5 minutes to make 5 widgets,
how long will it take 100 machines to make 100 widgets?

Step 1: Understanding

5 machines make 5 widgets in 5 minutes. We need to determine how long 100 machines will take to make 100 widgets.

Step 2: Reasoning

If 5 machines make 5 widgets in 5 minutes, then each machine makes 1 widget in 5 minutes.

Therefore, 100 machines working simultaneously will each make 1 widget in 5 minutes, producing 100 widgets total in 5 minutes.

Step 3: Calculation

- Rate of 1 machine = 1 widget per 5 minutes
- 100 machines = 100 widgets per 5 minutes
- Time for 100 widgets = 5 minutes

Step 4: Self-Check

- Original statement implies each machine produces 1 widget in 5 minutes.
- 100 machines produce 100 widgets in the same 5-minute period.
- The logic is consistent.

Final Answer: 5 minutes.



Screenshot 2: ***Bat and Ball Problem***

A bat and a ball cost \$1.10.
The bat costs \$1 more than the ball.
How much does the ball cost?

Step 1: Understanding

The total cost of a bat and a ball is \$1.10. The bat costs \$1 more than the ball.

Step 2: Reasoning

Let the ball cost x dollars.

Then the bat costs $x + 1$ dollars.

Together:

$$x + (x + 1) = 1.10$$

Step 3: Calculation

$$2x + 1 = 1.10$$

$$2x = 0.10$$

$$x = 0.05$$

So the ball costs \$0.05.

Step 4: Self-Check

- Ball = \$0.05
- Bat = \$1.05
- Total = \$0.05 + \$1.05 = \$1.10 ✓
- Difference = \$1.05 - \$0.05 = \$1.00 ✓

Final Answer: \$0.05 (5 cents)

Screenshot 3: *Lily Pads Problem*

Lily pads double every day.

A lake becomes completely covered in 48 days.

On which day was it half covered?

Step 1: Understanding

The lily pads double in area every day. The lake is completely covered on Day 48.

Step 2: Reasoning

If the coverage doubles each day, then one day before the lake is completely covered, it must have been half covered.

Step 3: Calculation

- Completely covered: Day 48
- Half covered: Day 47

Step 4: Self-Check

If the lake is half covered on Day 47 and the pads double on Day 48, the lake becomes fully covered. This is consistent with the problem.

Final Answer: Day 47.

6: Results Table

Test Case	Without CoT	With CoT
Machines Problem	Incorrect	correct
Bat and Ball	Incorrect	correct
Lily Pads	Incorrect	correct

7: Conclusion

The Chain-of-Thought prompt improved reasoning quality by forcing the model to solve problems systematically. The self-correction phase reduced logical errors and improved final answer accuracy.

8: Tools Used

- * ChatGpt*
- * Claude*
- * Google Docs*
- * PDF Export*